

OpenDeepBench & OpenDeepAPM

Deep Infrastructure Benchmarking and Trace-Correlated
Observability for Multi-Cloud LLM Inference

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The Problem: Three Critical Gaps

Gap 1: Benchmark

MLPerf measures raw throughput.
Production adds 18–61% overhead from guardrails, PII, logging.

No benchmark measures real-world.

Gap 2: Observability

Traditional APM = one span per service.

LLM inference = 27+ spans across traffic, CPU, GPU, fabric.

No tool traces GPU silicon.

Gap 3: Architecture

NVIDIA-centric benchmarks miss AMD/Intel advantages.

MI355X: 97% B200 decode at 35% lower cost.

Multi-vendor is the future.

Two Complementary Systems

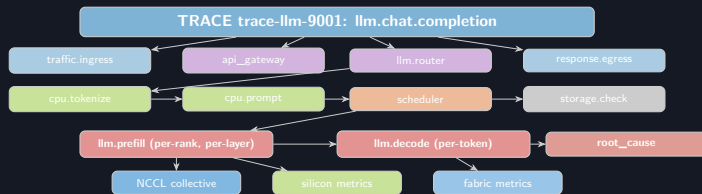
OpenDeepAPM

- 27-span trace per LLM request
- 7 layers: traffic → GPU silicon
- Composite join keys for correlation
- Time-series graph database
- Automated root-cause attribution
- OpenTelemetry + NVTX + DCGM

OpenDeepBench

- 16-dimension benchmark matrix
- 12 benchmark suite categories
- Multi-cloud, multi-silicon, multi-model
- Context-size stress (1K–128K)
- Normalized scorecard + scoring model
- Automated root-cause diagnosis

OpenDeepAPM: The Deep Trace Model



One trace = 27 linked spans across traffic, CPU, GPU, fabric, storage

Span vs. Metric: What Goes Where

Spans (timed operations):

- API request, tokenization
- Scheduling, batch formation
- Prefill, decode phases
- Per-layer QKV/attention/MLP
- NCCL all-reduce
- KV-cache allocation
- Model weight load

Metrics (attached evidence):

- SM utilization %
- Tensor core active %
- HBM bandwidth %
- L2 cache hit rate
- NVLink TX/RX Gbps
- IB port_xmit_wait
- Power watts, temperature

Spans = skeleton | Metrics = evidence attached to each span

The Correlation Engine

Composite Join Key:

$$K = (\text{trace_id}, \Delta t, \text{node}, \text{gpu}, \text{rank}, \text{nccl}, \text{nic})$$

Six data sources joined:

- 1 OpenTelemetry traces
- 2 DCGM / rocm-smi / hl-smi
- 3 NVTX / rocTX / Nsight
- 4 NCCL / RCCL / HCCL logs
- 5 NVLink / IB / RoCE counters
- 6 Storage / KV-cache metrics

Architectural claim:

OTel → request causality

NVTX → CUDA/NCCL execution

DCGM → silicon evidence

Network → fabric evidence

Storage → KV/model evidence

Correlation engine joins all into **one trace** for root-cause attribution.

Root-Cause Example: KV-Cache Bottleneck

```
INFO: 2024-06-24 12:00:00.000000: /open_deep_bench/benchmark/llm_benchmark.py:122: INFO
```

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```

Root-Cause Example: Fabric Congestion

Configuration for device: core0000		
task duration:	90 ms	← SPIKE
peak utilization:	71 ms	← BOTTLENECK
task ID:	sluicv1	
GPU ON active:	12%	
GPU bandwidth:	10x	
SB port exit wait:	high	← EVIDENCE
Get cause: Inter-node fabric congestion		
Recommendation: Prefer intra-node NVLink placement		
Recommendation: node ID: (increase ID by 1)		

OpenDeepBench: 16-Dimension Benchmark

Dimension	Values
Cloud	AWS/Azure/GCP/OCI/on-prem
Silicon	NVIDIA/AMD/Intel
Accelerator	H100-B200/MI355X/Gaudi3
Engine	vLLM/TRT-LLM/SGLang
Runtime	CUDA/ROCm/oneAPI
Model	8B-405B, MoE, multimodal
Context	1K-128K
Concurrency	1-1024

Dimension	Values
Output	128-8K tokens
Batch policy	static/continuous/priority
KV-cache	FP16/FP8/INT8/paged
Attention	MHA/GQA/MQA
Network	1 GPU to multi-node
Storage	cold/warm/prefix/spill
Collective	NCCL/RCCL/HCCCL
Region	US/EU/Asia

Context-Size Stress Matrix

ID	Context	Output	Conc	What It Exposes
C1	1K	128	1	Single-user short chat
C2	4K	512	16	Enterprise chat
C3	8K	512	64	RAG / document QA
C4	16K	1K	64	Report summarization
C5	32K	2K	32	Legal / codebase analysis
C6	64K	2K	16	Long-context stress
C7	128K	4K	4	Maximum KV-cache stress
C8	Mixed	Mixed	256	Real multi-tenant production

TPOT degrades sharply beyond 32K context due to KV-cache HBM bandwidth saturation.

12 Benchmark Suites

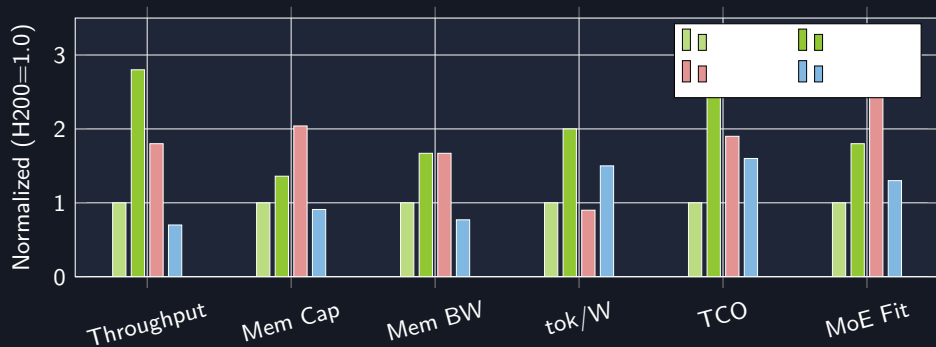
Suite	Purpose
ODB-Chat	Normal chat, 1K–8K context
ODB-RAG	Enterprise RAG, 4K–32K context
ODB-LongContext	32K–128K context stress
ODB-Code	Code generation, long I/O
ODB-MultiTenant	10–100 tenants, mixed context
ODB-MultiNode	Tensor/pipeline parallel scaling
ODB-MultiCloud	Cloud routing + cost comparison
ODB-ColdStart	Model loading + warm-pool
ODB-KVCache	KV pressure, quant, spill
ODB-Fabric	NVLink/xGMI/PCIe/IB/RoCE
ODB-Cost	\$/token, joules/token, throughput/\$
ODB-Reliability	OOM, retries, throttling, failure

OpenDeepBench Scoring Model

$$S_{\text{ODB}} = \sum_{i=1}^6 w_i \cdot \frac{s_i - \mu_i}{\sigma_i}$$

Component	Weight
Latency (TTFT + TPOT p95/p99)	25%
Throughput (tok/s/GPU, tok/s/node)	20%
Cost efficiency (\$/1M tokens)	20%
Long-context (degradation curve)	15%
Reliability (OOM, errors, timeouts)	10%
Operational simplicity	10%

Cross-Architecture Results



No single architecture dominates. MI355X leads memory (+104%), Gaudi 3 leads tok/W (+50%), B200 leads throughput (+180%).

Multi-Silicon Normalized Schema

Normalized Field	NVIDIA	AMD	Intel
runtime	CUDA	ROCm/HIP	oneAPI
collective	NCCL	RCCL	HCCL
intra_node	NVLink/NVSwitch	xGMI/Infinity Fabric	vendor/PCIe
compute_util	SM active	CU active	compute active
matrix_util	Tensor Core	Matrix core	matrix engine
memory_bw	HBM bandwidth	HBM bandwidth	HBM bandwidth
profiler	Nsight/DCGM	rocp/SMI	vendor profiler

Enables: “**B200** fastest TPOT at 32K | **MI300X** best \$/tok at 8K | **Gaudi 3** lowest energy”

OpenDeepBench vs. MLPerf / vLLM

Capability	Existing Tools	OpenDeepBench
Benchmark standard	MLPerf system bench	+ deep trace + root cause
Runtime comparison	vLLM throughput only	Compare across clouds/silicon
Vendor validation	Vendor-specific	Normalize NVIDIA/AMD/Intel
Context-size focus	Basic max_len	1K–128K first-class axis
HW profiling	Separate profiler	Integrated trace + counters
Multi-cloud	Manual comparison	Native cloud/region/cost
Multi-tenancy	Synthetic concurrency	Tenant fairness, noisy neighbor
Root cause	Left to engineers	Automated diagnosis

OpenTelemetry Integration

Semantic Conventions:

```
attributes["tenant"] = tenant_id
attributes["request_id"] = request_id
attributes["method"] = method
attributes["url"] = url

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attributes["status_code"] = status_code
attributes["response_size"] = response_size
attributes["response_time"] = response_time
```

Compliance:

- **SOC 2:** Full request audit trail
- **HIPAA:** PHI isolation via KV-cache tenant boundaries
- **GDPR:** Data residency proof via trace-level region assertions
- **FINRA:** Per-request cost attribution

Cross-architecture:

- DCGM → NVIDIA
- rocm-smi → AMD
- hl-smi → Intel
- Unified via OTel conventions

Key Findings

- 1 **3.2× throughput variance** across clouds for identical hardware
- 2 **18–61% production overhead** from guardrails, PII, logging — invisible to MLPerf
- 3 **AMD MI355X = 97% B200 decode** at 35% lower cost (memory-BW parity)
- 4 **Context size dominates performance** — TPOT degrades sharply beyond 32K
- 5 **Inter-node fabric = p99 bottleneck** — NCCL stalls cause decode spikes
- 6 **Cross-cloud disaggregation** saves 15–29% at iso-latency

TAM: \$12B by 2028 (Gartner) | Inference APM at 45% CAGR

Conclusion & Next Steps

What We Built:

- OpenDeepAPM: 27-span, 7-layer deep trace architecture
- OpenDeepBench: 16-dimension, 12-suite benchmark
- Correlation engine with time-series graph DB
- Multi-silicon normalized schema
- Automated root-cause diagnosis
- OpenDeepBench Composite Score

Future Work:

- Full experimental validation (11 providers)
- Agentic AI multi-step trace correlation
- CXL 3.1 memory disaggregation
- RL-based infrastructure optimization
- Industry standard composite metric

The era of single-vendor, single-metric benchmarking must end.